Stress Constraints

Use FEniCS, python and MMA to implement topology optimization of a problem including stress constraints. The implementation should use double filtering and be robust with respect to the stress constraints. The robust stress constraints should be implemented as “normal” constraints, e.g. for 3 design realizations

$$\min_{\boldsymbol{\rho}} f(\boldsymbol{\rho}) \;\; s.t.$$
$$\sigma_m^{Dialated} \leq f_\alpha \;\left(\sigma_y\right),$$
$$\sigma_m^{Nominal} \leq f_\alpha \;\left(\sigma_y\right),$$
$$\sigma_m^{Eroded} \leq f_\alpha \;\left(\sigma_y\right)$$
$$0 \leq \boldsymbol{\rho} \leq 1$$

or

$$\min_{\rho} \max \left\{f^{Dialated}(\boldsymbol{\rho}), f^{Nominal}(\boldsymbol{\rho}), f^{Eroded}(\boldsymbol{\rho})\right\} \;\; s.t.$$
$$\sigma_m^{Dialated} \leq f_\alpha \;\left(\sigma_y\right),$$
$$\sigma_m^{Nominal} \leq f_\alpha \;\left(\sigma_y\right),$$
$$\sigma_m^{Eroded} \leq f_\alpha \;\left(\sigma_y\right)$$
$$0 \leq \boldsymbol{\rho} \leq 1$$

See lecture 12 slide 13 for how to implement the stress constraints. Start without adaptive normalization and add it only if you think it is needed.

Realize your final geometry in a CAD program and perform a FEM calculation using the CAD geometry in your favorite FEM program (FEniCS? 😊).
Compare the stresses found in the CAD geometry to those in the final iteration of the topology optimization.
Check if the stress constraints are violated in the calculation based on the CAD geometry.